Final year project proposal

SCAN2LEARN AI

AI-POWERED SMART SCANNING AND LEARNING ASSISTANT

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1. Project Abstract

Scan2Learn AI is an AI-powered Android application designed to help students convert printed educational materials into digital and interactive learning content. The application allows users to capture a picture of any printed page using mobile camera or upload an existing image. Using Optical Career Recognition (OCR) the system extracts text from the image and display it in to a readable form.

The application uses Artificial Intelligence (AI) to generate summaries, multi –choice questions (MCQs), and answers to question related to the scanned content. It also provide a Text-to Speech (TTS) feature that allows users to listen to extracted text in supported languages. The application will be developed using Flutter and Dart, with a secure backend for AI API integration and Firebase for storing user data and learning history.

The main purpose of Scan2Learn AI is make studying easier by combining text extraction, AI assisted content understanding, question generation, and voice output in a single Android application.

2. Problem Statement

Students offer study from printed textbooks, lecture notes, and handouts but may need multiple applications to scan pages, extract text, generate summaries and MCQs, ask questions, and listen to content. Although AI tools like ChatGPT provide these features, image –upload limits may interrupt the learning process when students need to analyze multiple pages.

Switching between different applications can also be time-consuming and inconvenient. Scan2Learn AI aims to solve this problem by combining documents scanning, text extraction, AI-generated summaries and MCQs, content-based question answering, and text-to-speech in a single Android application.

3. Key Objectives:

OCR Text Extraction:

Extract text from page images using Google ML Kit.

AI Summarization:

Generate concise summaries from extracted text using Claude API.

MCQs Generation:

Automatically-choice questions with four options and correct answers.

AI Questions Answering:

Answer questions based on the scanned educational content.

Text-to Speech:

Read extracted text and summaries aloud in supported languages.

Fired base Integration:

Save scanned content and generated learning material.

Security:

Protect user data and API credentials through secure backend communication.

4. Technical Architecture & Stack

Fronted:

Flutter and Dart (Android application).

OCR Engine:

Google ML Kit Text Recognition.

Backend API:

Firebase Cloud Functions.

AI Engine:

Claude API for summaries, MCQs and question answering.

Database:

Firebase Cloud Fire store.

Voice Output:

Android- compatible Text-to-Speech (TTS).

Development Tools:

Visual Studio Code.

5. High-level-Database Schema

The Proposed system will use firebase cloud firestore to store and manage the information required for user accounts and saved learning content.

Users: User ID, email, account details.

Scanned Documents:

Documents ID, extract text, image reference.

Generated Content:

Summary, MCQs, answers, creation date.

Learning History:

User ID, document ID, last accessed date.

This structure will allow users to save and retrieve their learning materials securely.

6. Timeline

Phase 1(weeks 1-4)

- Requirement analysis and project planning.
- UI wireframes and database design.
- Flutter application setup.

Phase 2 (Week 5-8)

- Google ML Kit OCR integration.
- Claude API integration.
- Summary, MCQs generation, and question-answering features.

Phase 3 (week 9-12)

- Firebase integration:
- Text-to-Speech implementation.
- Security checks, testing and documentation.

7. Financial Feasibility:

Development Cost:

Expected to be PKR 0 using free development tools.

OCR Cost:

Expected to be free per request when using on-device ML Kit recognition.

Database and Backend:

May remain within available free-tier limits.

AI API Cost:

May incur usage-based charges depending on the number of request.

Total Project Cost:

Depends on final service selection and usage.

The project is financially manageable as an academic prototype by using free tools and controlling API usage.